

VisionBoard

Real-Time PCB Defect Detection Using YOLO-Based AI Vision

Transforming PCB quality control through intelligent, agentic defect detection

PROBLEM STATEMENT

PCB manufacturing often produces defective boards, solder bridges, missing components, and misaligned parts that human inspectors regularly miss. This leads to wasted boards, higher costs, and products that fail in the customer's hands. We are building an AI-powered solution using YOLO to automatically detect these defects in real time during manufacturing, before they become a bigger problem

PROBLEM DESCRIPTION

PCB production is a precise process where tiny errors can ruin an entire board. Defects can happen at any stage from applying solder paste to placing components to final assembly. Current inspection methods are either done by hand, by outdated machines with fixed rules, or only checked at the very end when it's too late. None of these catch defects early enough or consistently enough.:

Current Method	Why It Falls Short
Manual Visual	Fatigue-prone, inconsistent, misses sub-mm defects
Rule-Based Automated Optical Inspection	High false-positive rate, brittle to design changes
Post-Production Testing	Defects caught too late rework is costly

We use a YOLO AI model connected to cameras on the PCB production line. It scans each board in real time, automatically spots defects like solder bridges and missing components, and flags faulty boards instantly. Engineers see all of this on a live dashboard so they can act fast.

RATIONALE

Solving this problem with AI native engineering delivers measurable value across quality, cost, and speed:

Defect Escape Rate AI detection catches defects human inspectors routinely miss, significantly reducing escape rate throughout production.	Scrap & Rework Cost Early detection means fewer boards reach late-stage assembly with faults, directly cutting rework and scrap costs.	Inspection Speed YOLO processes frames in milliseconds inspection becomes a non-bottleneck on the production line.	Customer Satisfaction Higher yield and consistent quality translate directly into reliable products and stronger customer trust.
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Vision: Build the first fully AI-native PCB inspection platform where defect detection, classification, reporting, and process feedback operate autonomously empowering manufacturers to achieve near-zero defect output.

Group Name : VisionX

